



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 06

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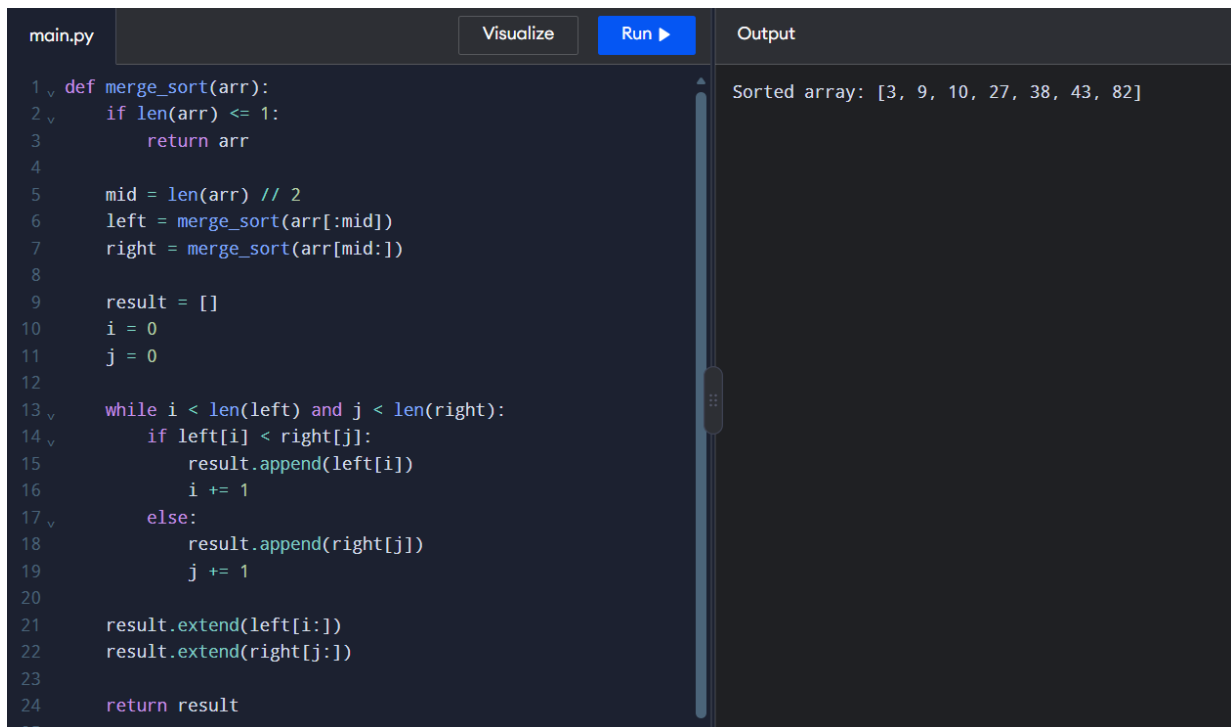
Reg No: 192525020

Program:

```
def merge_sort(arr):  
    if len(arr) <= 1:  
        return arr  
    mid = len(arr) // 2  
    left = merge_sort(arr[:mid])  
    right = merge_sort(arr[mid:])  
    result = []  
    i = 0  
    j = 0  
    while i < len(left) and j < len(right):  
        if left[i] < right[j]:  
            result.append(left[i])  
            i += 1  
        else:  
            result.append(right[j])  
            j += 1  
    result.extend(left[i:])  
    result.extend(right[j:])  
    return result  
arr = [38, 27, 43, 3, 9, 82, 10]
```

```
print("Sorted array:", merge_sort(arr))
```

Output:



The image shows a code editor interface with a dark theme. On the left, a file named 'main.py' is open. The code is a Python implementation of the merge sort algorithm. It includes a function definition, a base case, a recursive splitting phase, a merging phase with a while loop, and a final return statement. On the right, there is an 'Output' panel showing the result of running the code. Above the code editor, there are buttons for 'Visualize' and 'Run'. A vertical scrollbar is visible between the code and output panels.

```
1 def merge_sort(arr):
2     if len(arr) <= 1:
3         return arr
4
5     mid = len(arr) // 2
6     left = merge_sort(arr[:mid])
7     right = merge_sort(arr[mid:])
8
9     result = []
10    i = 0
11    j = 0
12
13    while i < len(left) and j < len(right):
14        if left[i] < right[j]:
15            result.append(left[i])
16            i += 1
17        else:
18            result.append(right[j])
19            j += 1
20
21    result.extend(left[i:])
22    result.extend(right[j:])
23
24    return result
```

Sorted array: [3, 9, 10, 27, 38, 43, 82]